The Terrain-Following One-Way Equation as a Pseudodifferential Problem

Note accompanying the TAP 2026 manuscript (`main.tex`)

Abstract

The Liouville-transformed equation $\varphi_{\xi\xi} + B\varphi = 0$ (equation (11) of the manuscript) is read in the language of pseudodifferential operators: ξ is the evolution variable and B is an operator in η with ξ-dependent coefficients.

- We compute the exact Weyl symbol of B. Its principal part is the local dispersion relation, and it has no first-order part. The only corrections are two zeroth-order terms of relative size $(\lambda/\ell)^2$: the potential V of the manuscript, and a term $r_\eta^2/(4r^4)$ that the operator form keeps implicitly.
- The one-way factorization (13) is the symbol expansion of the operator Riccati equation. We give its next term, which reduces to the classical second-order WKB correction.
- The Bremmer coupling series makes the neglected backward wave explicit. Its local part reproduces the second-order term. Genuine backscatter is $O(\varepsilon^\infty)$, exponentially small for smoothed terrain. The flux pairs are the interface transmission coefficients without the staircase reflection loss.
- The flux-conserving step, the rational propagator, the per-class reference wavenumber, the dispersion analysis, the vertical stretching and the nonuniform range step are then statements about symbols.
- The same calculus explains why the shift map has an error floor that does not vanish under refinement, and why no local top boundary absorbs grazing waves.

1 Setting and conventions

The equation. In orthogonal coordinates (ξ, η) with scale factors h_ξ, h_η and anisotropy $r = h_\eta/h_\xi$, the Helmholtz equation reads (manuscript (7))

$$\partial_\xi(r\partial_\xi\psi) + \partial_\eta(r^{-1}\partial_\eta\psi) + h_\xi h_\eta k^2\psi = 0. \tag{1}$$

With $\psi = r^{-1/2}\varphi$ it becomes (manuscript (11))

$$\varphi_{\xi\xi} + B\varphi = 0, \qquad B\varphi = r^{-1/2}\partial_\eta\Big(r^{-1}\partial_\eta(r^{-1/2}\varphi)\Big) + (k^2h_\xi^2 - V)\varphi, \tag{2}$$

with $V = (r^{1/2})_{\xi\xi}/r^{1/2}$.

Quantization. Let $D = -i\partial_\eta$ and let q be the variable dual to η; q is the computational vertical wavenumber of Section VII of the manuscript. For a function $a(\eta, q)$, depending on ξ as a parameter, the Weyl quantization is

$$\mathrm{Op}^w(a)u(\eta) = \frac{1}{2\pi}\iint e^{i(\eta-\eta')q}\, a\Big(\tfrac{\eta+\eta'}{2}, q\Big)\, u(\eta')\, d\eta'\, dq.$$

Real symbols give symmetric operators. Composition is given by the Moyal product

$$a\#b = ab + \frac{i}{2}\{a,b\} - \frac{1}{8}\left(a_{qq}b_{\eta\eta} - 2a_{q\eta}b_{q\eta} + a_{\eta\eta}b_{qq}\right) + \ldots, \qquad \{a,b\} = a_\eta b_q - a_q b_\eta. \tag{3}$$

Odd-order terms are antisymmetric in (a,b). Hence $a\#a = a^2 + O(\partial^4)$, and the symbol of a commutator is $i\{a,b\}$ to leading order.

The small parameter. Let ℓ be the shortest length on which the metric varies: the smoothing length of the terrain, or the transition height L. The expansion parameter is $\varepsilon = 1/(k\ell)$. In rescaled variables η/ℓ and q/k, every derivative of a coefficient brings one factor ε; this is the semiclassical calculus with $\hbar = \varepsilon$ [1, 2]. Symbols are graded by powers of ε. We write $a = a_0 + \varepsilon a_1 + \varepsilon^2 a_2 + \ldots$ but keep the physical variables in the formulas.

2 The symbol of B

Exact Weyl symbol. For any smooth g we have $\mathrm{Op}^w(g\,q^2) = \frac{1}{4}(D^2 g + 2DgD + gD^2)$. Moreover $D^2 g + gD^2 = 2DgD - g''$, so

$$DgD = \mathrm{Op}^w\big(g\,q^2 + \tfrac{1}{4}g''\big)\,. \tag{4}$$

With $f = r^{-1/2}$ and $g = r^{-1}$, moving f through D gives $fDgDf = D(f^2 g)D - (ff'g)' + gf'^2$. The derivative part of B is $-fDgDf$. Using (4) for $f^2 g = r^{-2}$, the second derivatives of r cancel and only a square of r_η is left:

$$\boxed{b(\xi,\eta,q) = k^2 h_\xi^2 - \frac{q^2}{r^2} - V - \frac{r_\eta^2}{4r^4}}, \qquad B = \mathrm{Op}^w(b) \text{ exactly.} \tag{5}$$

The identity $r^{-1/2}(r^{-1}(r^{-1/2}u)')' = (r^{-2}u')' + \frac{1}{4}(r^{-2})''u - \frac{1}{4}r'^2 r^{-4}u$ behind (5) was checked numerically to working precision.

Principal symbol: the dispersion relation. The principal symbol is

$$b_0 = k^2 h_\xi^2 - \frac{q^2}{r^2}.$$

Let κ be dual to ξ. The characteristic set of (2) is $\kappa^2 = b_0$, that is, $(\kappa/h_\xi)^2 + (q/h_\eta)^2 = k^2$: the local plane waves of the manuscript. With $q = kh_\eta \sin\theta$, where θ is the angle to the level curve,

$$b_0 = k^2 h_\xi^2 \cos^2\theta, \qquad \sqrt{b_0} = kh_\xi \cos\theta. \tag{6}$$

No first-order part. The symbol (5) is real, with nothing of order ε^1. This is the symbol form of "B is symmetric in $L^2(d\eta)$". It matters: a first-order (subprincipal) term would be an imaginary part of order εq, which transports amplitude across η and makes B non-normal. The original equation (1), divided by r, has first-order terms in both variables: $(r_\xi/r)\partial_\xi$ and $(\partial_\eta r^{-1})/r^{-1}\cdot\partial_\eta$. The Liouville substitution is the gauge that removes them. The price is the two zeroth-order terms of order ε^2:

- $V = (r^{1/2})_{\xi\xi}/r^{1/2}$ from the ξ direction. Since ξ is the evolution variable and is not quantized, this term has to be written explicitly, and the scheme computes it by a second difference along the columns.

- $W = r_\eta^2/(4r^4)$ from the η direction. It is not written in the manuscript and does not need to be: the scheme discretizes the operator form $r\partial_\eta r^{-1}\partial_\eta$, which contains W exactly.

Both are $O((\lambda/\ell)^2)$ relative to $k^2 h_\xi^2$, which is the size stated after (11) for V.

The vertical stretching. A stretching $\eta = Z(\eta')$ with $s = dZ/d\eta'$ multiplies h_η by s. In (5) this replaces r by rs, so $q'/(h_\eta s)$ is the physical vertical wavenumber. This is the symbol form of "a stretching is a metric" (`nonuniform_grids.md`, Section 1).

3 One-way factorization as a symbol expansion

The Riccati equation. Seek $\varphi_\xi = i\Lambda\varphi$ with $\Lambda = \mathrm{Op}^w(\lambda)$. Then (2) requires

$$\Lambda^2 - i\Lambda_\xi = B \quad \Longleftrightarrow \quad \lambda\#\lambda - i\partial_\xi\lambda = b. \tag{7}$$

We expand $\lambda = \lambda_0 + \lambda_1 + \lambda_2 + \dots$, with λ_j of order ε^j relative to λ_0, and ∂_ξ of a symbol counts as one order.

Order 0. $\lambda_0^2 = b_0$. The outgoing root is $\lambda_0 = \sqrt{b_0}$, with $\mathrm{Im}\,\sqrt{b_0} \geq 0$; for evanescent waves ($b_0 < 0$) it decays in ξ.

Order 1. The first Moyal term of $\lambda\#\lambda$ vanishes, so $2\lambda_0\lambda_1 - i\partial_\xi\lambda_0 = 0$ and

$$\lambda_1 = \frac{i}{2}\frac{\partial_\xi\lambda_0}{\lambda_0} = \frac{i}{4}\frac{\partial_\xi b}{b}. \tag{8}$$

This is the term $\frac{i}{4}B^{-1}B_\xi$ of (13) in the manuscript, and the manuscript's "treated as commuting" is exact at this order. Indeed, $\frac{i}{4}B^{-1}B_\xi$ has the symbol $\frac{i}{4}[b_\xi/b + \frac{i}{2}\{b^{-1}, b_\xi\} + \dots]$, whose correction is of order ε^2. The operator ordering only matters from the next order on.

Order 2. Collecting the terms of order ε^2 in (7), including the second Moyal term of $\lambda_0\#\lambda_0$, gives

$$\lambda_2 = \frac{1}{2\sqrt{b}}\left[\frac{5\,b_\xi^2}{16\,b^2} - \frac{b_{\xi\xi}}{4b} + \frac{1}{4}\left(\partial_q^2\lambda_0\,\partial_\eta^2\lambda_0 - (\partial_q\partial_\eta\lambda_0)^2\right)\right]. \tag{9}$$

- If b does not depend on η, the last bracket vanishes. What remains is the classical second-order WKB correction $\frac{5}{32}Q'^2Q^{-5/2} - \frac{1}{8}Q''Q^{-3/2}$ for $y'' + Qy = 0$: a check of the expansion.
- The bracket is the phase-space curvature of the one-way symbol. Together with the ordering ambiguity of $\frac{i}{4}B^{-1}B_\xi$ (a term $\propto \{b^{-1}, b_\xi\}$), it is what the manuscript calls "second-order terms and commutators $[B, B_\xi]$", whose symbol is $i\{b, b_\xi\}$.
- All of these are $O(\varepsilon^2)$. Over a transition region of length ℓ_ξ they accumulate a phase of order $k\ell_\xi\varepsilon^2$.

Limits of the expansion.

- *Smoothness.* The expansion needs $\varepsilon \ll 1$. The terrain smoothing and the transition height are what keep the metric smooth on the wavelength scale.
- *Turning points.* The expansion fails where $b_0 = 0$, i.e. $\theta \to 90°$ to the marching direction, because $\sqrt{b_0}$ is not smooth there. Backscatter is not captured; this is the general limit of one-way methods [4, 5]. Section 5 quantifies it. Within the aperture, $b_0 \geq k^2 h_\xi^2 \cos^2\theta_{\max} > 0$ keeps the field away from it.

4 Flux and the factors $B^{\pm 1/4}$

Flux. The flux $\mathcal{F} = \mathrm{Re}\langle\varphi, \sqrt{B}\varphi\rangle$ is the quadratic form with symbol $\sqrt{b_0}$, by (14) of the manuscript. By (6) it is the squared amplitude times $kh_\xi \cos\theta$, which is the WKB amplitude law of the manuscript.

Conjugation. The variable $\chi = B^{1/4}\varphi$ obeys

$$\chi_\xi = B^{1/4}\left(i\sqrt{B} - \tfrac{1}{4}B^{-1}B_\xi\right)\varphi + \left(B^{1/4}\right)_\xi \varphi = i\sqrt{B}\chi + O(\varepsilon^2),$$

because $(B^{1/4})_\xi$ has the symbol $\frac{1}{4}b^{-3/4}b_\xi$ up to $O(\varepsilon^2)$. The conjugation removes λ_1 exactly. So the transport term is a gauge of order ε, and the flux-conserving step is a gauge-corrected exponential.

The flux pairs. In (15) of the manuscript the middle factor $e^{i\Delta\xi\sqrt{B_{n+1/2}}}$ commutes exactly with $B_{n+1/2}^{\pm 1/4}$: all three are functions of one operator, and for symbols $\{F(b), G(b)\} = F'G'\{b, b\} = 0$. The two brackets have the symbols

$$\left(\frac{b_{n+1/2}}{b_{n+1}}\right)^{1/4} = 1 - \frac{\Delta\xi}{8}\frac{\partial_\xi b}{b} + O(\Delta\xi^2), \qquad \left(\frac{b_n}{b_{n+1/2}}\right)^{1/4} = 1 - \frac{\Delta\xi}{8}\frac{\partial_\xi b}{b} + O(\Delta\xi^2).$$

- They are the identity wherever the two columns coincide, in particular in the exterior.
- Their composition with the exponential is exact up to Poisson brackets between neighbouring columns, $\{b_n, b_{n+1/2}\} = \frac{\Delta\xi}{2}\{b, \partial_\xi b\} + \dots$. These are the neglected commutators of the previous section, now per step.

The rotated Padé factors. Milinazzo's rotation of the factors $(1+L)^{\pm 1/4}$ moves their poles off the evanescent part of the symbol range $b < 0$. A pair of factors at slightly different columns, whose symbols almost cancel, then no longer has a pole at almost the same real point, which is where the spurious amplification arose.

5 The Bremmer coupling series

The one-way equation drops the backward wave. The Bremmer series [6, 7, 5] keeps both waves and treats their coupling as a perturbation. It shows what the one-way step neglects, and how large that is.

Splitting. Let $B > 0$ on the propagating part of the spectrum, with $\Lambda = B^{1/2}$ and $Q = B^{1/4}$, so that $Q^2 = \Lambda$ and $Q^{-1}BQ^{-1} = \Lambda$ exactly. Define the flux-normalized amplitudes $u_\pm$ by

$$\varphi = Q^{-1}(u_+ + u_-), \qquad \varphi_\xi = iQ(u_+ - u_-), \qquad u_\pm = \tfrac{1}{2}\left(Q\varphi \mp iQ^{-1}\varphi_\xi\right). \tag{10}$$

Substituting into (2) gives, with no approximation,

$$\partial_\xi \begin{pmatrix} u_+ \\ u_- \end{pmatrix} = \begin{pmatrix} i\Lambda + T & \mathcal{R} \\ \mathcal{R} & -i\Lambda + T \end{pmatrix} \begin{pmatrix} u_+ \\ u_- \end{pmatrix}, \qquad T = \tfrac{1}{2}\left[Q_\xi, Q^{-1}\right], \quad \mathcal{R} = \tfrac{1}{2}\left(Q_\xi Q^{-1} + Q^{-1}Q_\xi\right). \tag{11}$$

- Q_ξ and Q^{-1} are symmetric, so T is antisymmetric and $\mathcal{R}$ is symmetric. Hence $\partial_\xi\left(\|u_+\|^2 - \|u_-\|^2\right) = 0$: the coupled system conserves the flux exactly, and $\|u_\pm\|^2$ are the forward and backward fluxes.

- The symbols follow from the Moyal product (3), whose first-order term cancels in the anti-commutator and survives in the commutator:

$$\rho := \text{symbol}(\mathcal{R}) = \frac{\partial_\xi b}{4b}\big(1 + O(\varepsilon^2)\big), \qquad \text{symbol}(T) = \frac{i}{32}\frac{\{b, \partial_\xi b\}}{b^2} + O(\varepsilon^3). \tag{12}$$

  Relative to $\lambda_0 = \sqrt{b}$, the coupling ρ is of order ε and T of order ε^2. T is the commutator $[B, B_\xi]$ that the manuscript neglects, and it is a pure phase.

- In the exterior, and wherever the columns do not change, $\rho = 0$ and $T = 0$, and the two waves decouple exactly.

The series. Let $G_\pm(\xi, \xi')$ be the propagators of $\partial_\xi \mp i\Lambda - T$. For flux-normalized variables they are unitary. With the incident field given at $\xi = 0$ and no backward wave entering from the far end $\xi = X$,

$$u_+(\xi) = G_+(\xi, 0)u_+(0) + \int_0^\xi G_+(\xi, \xi')\mathcal{R}u_-(\xi')\, d\xi', \qquad u_-(\xi) = -\int_\xi^X G_-(\xi, \xi')\mathcal{R}u_+(\xi')\, d\xi'.$$

Iterating, starting from $u_- = 0$, gives the Bremmer series. Its terms alternate between forward and backward propagation, with one scattering by $\mathcal{R}$ at each turn.

- The zeroth term $u_+ = G_+u_+(0)$ is the one-way equation in the variable $\chi = B^{1/4}\varphi$ of the previous section. When $u_- = 0$, (10) gives $u_+ = Q\varphi = \chi$ exactly. The flux-conserving step of the manuscript is the zeroth Bremmer term, with T dropped.

- Because $G_\pm$ are unitary, the n-th term is bounded by $\left(\int_0^X \|\mathcal{R}\|\, d\xi\right)^n$. The series therefore converges geometrically when $\int \|\mathcal{R}\| d\xi < 1$. By (12) this integral is $\frac{1}{2}$ of the total variation of $\log(h_\xi \cos\theta)$ along the path, which is small for the smoothed terrains of the manuscript as long as $\cos\theta$ stays away from 0.

Two parts of the first reflection. Take a forward wave $u_+ \approx e^{i\Phi}$, with $\Phi_\xi = \lambda_0$, and consider one column first. The first backward term is

$$u_-(\xi) = -e^{-i\Phi(\xi)} \int_\xi^X \rho(\xi')\, e^{2i\Phi(\xi')}\, d\xi'. \tag{13}$$

Integration by parts splits it into a local part and a remainder:

$$u_-(\xi) = \underbrace{\frac{\rho}{2i\lambda_0}\, e^{i\Phi(\xi)}}_{\text{slaved, local}} + \underbrace{e^{-i\Phi(\xi)} \int_\xi^X \left(\frac{\rho}{2i\lambda_0}\right)_{\xi'} e^{2i\Phi(\xi')}\, d\xi'}_{\text{genuine backscatter}}.$$

- *The slaved part* travels forward with the wave: its phase is $e^{+i\Phi}$, not $e^{-i\Phi}$. It is a local dressing of size ε that vanishes where the metric stops changing, and it is not a reflected wave. Fed back into the u_+ equation through $\mathcal{R}$, it changes the forward rate by $\rho \cdot \rho/(2i\lambda_0)$, so the effective one-way symbol becomes $\lambda_0 - \rho^2/(2\lambda_0)$. Undoing the transformation (10) adds the gauge term $-\partial_\xi(\rho/2\lambda_0)$. With $\rho = b_\xi/(4b)$ this gives

$$-\frac{\rho^2}{2\lambda_0} - \partial_\xi \frac{\rho}{2\lambda_0} = \frac{5\, b_\xi^2}{32\, b^{5/2}} - \frac{b_{\xi\xi}}{8\, b^{3/2}},$$

which is exactly λ_2 of (9) without η dependence. So the second-order terms of the one-way expansion are the near-diagonal part of the double reflection. They are local and of order ε^2, and they can be added to the one-way operator without any backward propagation.

- *The genuine backscatter* is an oscillatory integral with phase 2Φ and $\Phi_\xi = \lambda_0 \geq kh_\xi \cos\theta_{\max} > 0$. It has no stationary point. Each further integration by parts gains a factor ε, so it is $O(\varepsilon^\infty)$ for a smooth metric. If the metric is analytic in a strip of width δ around the real ξ axis, it is exponentially small, about $e^{-2kh_\xi \cos\theta\,\delta}$.

Check. For $y''+b(x)y=0$ with $b=k^2(1+a\tanh(x/\ell))$ and $a=0.3$, the exact reflection coefficient and the first Bremmer term $\left|\int \rho\, e^{2i\Phi} dx\right|$ are:

$k\ell$	0.5	1	2	3
$\lvert R\rvert$ exact	$1.07\cdot10^{-1}$	$4.45\cdot10^{-2}$	$4.44\cdot10^{-3}$	$3.55\cdot10^{-4}$
first Bremmer term	$1.07\cdot10^{-1}$	$4.43\cdot10^{-2}$	$4.41\cdot10^{-3}$	$3.53\cdot10^{-4}$

The two agree to within 0.6%. The decay is exponential in $k\ell$, at about the rate $e^{-\pi k_{\min}\ell}$ set by the poles of tanh at distance $\pi\ell/2$ from the real axis.

Multidimensional picture. In phase space, $\mathcal{R}$ couples the forward sheet $\kappa = +\lambda_0(\eta,q)$ to the backward sheet $\kappa = -\lambda_0(\eta,q)$. Within the aperture the two sheets are separated by $2\lambda_0 \geq 2kh_-\cos\theta_{\max}$. An operator with a smooth symbol cannot transfer energy between disjoint characteristic sets beyond $O(\varepsilon^\infty)$, which is the microlocal form of the non-stationary phase above. The sheets meet only where $b_0 = 0$. There $\rho = b_\xi/(4b)$ and $1/\lambda_0$ blow up, and the series stops being a perturbation series. This is the turning-point limit of the one-way method again, now as a statement about coupling.

What the one-way step neglects. The Bremmer series ranks the neglected terms:

1. the phase T and the slaved coupling, $O(\varepsilon^2)$, both local. These are the second-order terms and commutators named in the manuscript, and they could be included in a one-way step;
2. genuine backscatter: $O(\varepsilon^\infty)$, exponentially small for analytic (for example Gaussian-smoothed) terrain.

The smoothing of the terrain is therefore not just a numerical convenience. A kink in $T(x)$ makes the metric non-analytic. The backscatter then decays only algebraically in k: it is edge diffraction, which no one-way method represents. This is the generalized Bremmer series of de Hoop [5]. Choosing Λ to include higher terms of the Riccati expansion pushes $\mathcal{R}$ to higher order, and in the limit the forward and backward waves are decoupled to all orders.

The flux pairs as interface transmission. The column-by-column step replaces the continuous metric by a staircase in ξ. At an interface between columns with roots λ_a and $\lambda_b = \lambda_a(1+\delta)$, continuity of φ and φ_ξ gives the amplitude transmission $t = 2\lambda_a/(\lambda_a+\lambda_b)$ and reflection $\delta/(2+\delta)$. In flux-normalized form,

$$t = \left(\frac{\lambda_a}{\lambda_b}\right)^{1/2} \tau = \left(\frac{b_a}{b_b}\right)^{1/4} \tau, \qquad \tau = \frac{2\sqrt{\lambda_a\lambda_b}}{\lambda_a+\lambda_b} = 1 - \frac{\delta^2}{8} + O(\delta^3),$$

with $\tau^2 + |\delta/(2+\delta)|^2 = 1$.

- The flux pairs of (15) are exactly the factor $(b_a/b_b)^{1/4}$, so the step is the staircase transmission with τ set to 1.
- This is the better choice. Here $\delta \approx 2\rho\,\Delta\xi$, so each interface reflects a flux of about $\rho^2\Delta\xi^2$. Over a length ℓ_ξ the staircase would remove a forward flux of order $\rho^2\ell_\xi\Delta\xi$ through backscatter. That loss is an artefact of the staircase: it vanishes as $\Delta\xi \to 0$, and the reflections of the continuous metric cancel by phase as shown above.
- Taking $\tau = 1$ removes the artefact and conserves the forward flux exactly. This is the discrete counterpart of the zeroth Bremmer term.

Caveat: evanescent part. For $b < 0$ the roots Λ and Q are complex and T, $\mathcal{R}$ lose their (anti)symmetry. The splitting (11) still holds algebraically with the decaying branch, but it no longer carries a flux interpretation. Evanescent components do not propagate, so the estimates above concern the propagating sector.

Can the existing JAX propagators implement this series? Yes, as a computational framework, but not by simply running the existing one-way solver forward and backward. The repository's `RationalHelmholtzPropagator` and `SSFHelmholtzPropagator` advance a single reduced field in range; their `compute` methods do not expose the transverse Helmholtz operator B, its square root, or a backward amplitude. In particular, reversing the range loop is not the inverse of a lossy, terrain-masked step and would not impose the two-end boundary conditions in the Bremmer integral equations.

The reusable parts are the JAX arrays, terrain and heterogeneity sampling, vertical discretizations, boundary-condition machinery, and compiled range loops. A first implementation for the existing flat-coordinate 2D problem can form a discrete transverse operator B_j at each range column using the same vertical stencil (rational solver) or spectral multiplier (SSF solver). On its propagating subspace, apply $B_j^{1/2}$ and $B_j^{1/4}$ with the existing spectral representation where available, or a rational/Krylov approximation for the finite-difference operator. Then compute the discrete coupling from the change in $Q_j = B_j^{1/4}$, preferably as the symmetric formula in (11); this avoids differentiating a noisy terrain profile. JAX's `lax.scan`/`fori_loop` can march u_+ from the source and u_- from the far-range no-incoming-wave boundary, and repeat the two sweeps to sum a chosen number of Bremmer reflections. The far boundary and all range-dependent terrain masks must be included in this two-point iteration rather than inherited from the one-way absorber.

This is a moderate new solver layer, not a small option on `compute`. The most direct prototype is SSF with a flat, lossless domain and a smooth range-varying medium: its vertical spectrum makes the roots and flux normalization accessible. The rational finite-difference solver can follow once a stable matrix-function action and compatible flux form are specified. The current top absorbing layer and lossy/complex media also need separate care, since the unitary $G_\pm$ and conserved flux used in the convergence bound apply only to the self-adjoint propagating problem. Validate first against a range-dependent 1D profile with an exact reflection coefficient, then compare the zeroth and first Bremmer orders against the existing one-way result. Thus the existing JAX infrastructure is suitable, but a faithful two-way implementation requires explicit discrete factorization and a two-boundary iteration.

6 The rational step: functional calculus

Symbol of the step. For $B = \mathrm{Op}^w(b)$ symmetric and a smooth function F, the semiclassical functional calculus gives

$$F(B) = \mathrm{Op}^w(F \circ b) + O(\varepsilon^2) \tag{14}$$

[3, 2]. The $O(\varepsilon)$ correction vanishes in the Weyl calculus. Apply this to $F = R$, the rational step (3), in the variable $L = B/\beta^2 - 1$:

$$\mathrm{symbol}\big(R(L)\big) = R\Big(\frac{b}{\beta^2} - 1\Big) + O(\varepsilon^2), \qquad \mathrm{symbol}\big(e^{i\beta\Delta\xi(\sqrt{1+L}-1)}\big) = e^{i\Delta\xi(\sqrt{b}-\beta)} + O(\varepsilon^2).$$

So the error of the rational approximation is, to leading order, a pointwise function on phase space,

$$E(\xi, \eta, q) = \Big| R\Big(\tfrac{b}{\beta^2} - 1\Big) - e^{i\Delta\xi(\sqrt{b}-\beta)} \Big|.$$

The error budget. What has to be small is E on the support of the field in phase space: its wavefront set, or in practice the support of its Wigner function. By (6), for angles up to the aperture $\theta_{\max}$ the symbol b ranges over $[k_{\min}^2 h_-^2 \cos^2\theta_{\max},\, k_{\max}^2 h_+^2]$, with $h_\pm = \max / \min h_\xi$. These are the rescaled bounds with which the grid optimizer is called (Section VII of the manuscript). The "local equivalence" is the statement that E depends on the metric only through this range.

Per-class reference wavenumber. On one column, b takes values only in that column's h_ξ range. With β_k chosen per class, the Padé approximant has to cover a narrower interval of $b/\beta_k^2 - 1$ (`beta_mode='column'`). The envelope factor $e^{i(\beta_k-\beta)\Delta\xi}$ is a constant phase and does not change the symbol.

Where the field lives in phase space: bicharacteristics. The support of the field is not a property of one column. It is transported by the Hamiltonian flow of the one-way symbol,

$$\frac{d\eta}{d\xi} = -\partial_q \sqrt{b_0}, \qquad \frac{dq}{d\xi} = \partial_\eta \sqrt{b_0}, \tag{15}$$

and reflected at the ground by the boundary symbol (manuscript (21)).

- A ray reflected by a slope θ_s leaves with up to $\theta_{\max} + 2\theta_s$ to the horizontal.
- Once the terrain flattens, its local symbol range is that of a flat column, but its q is still large.
- This is why the nonuniform range step (`range_steps='class'`) failed when the aperture did not include the slopes. Each class's step was sized for the q range of its own columns, while the field there carried the larger q produced upstream. The error budget has to be evaluated on the transported support, i.e. with the slope-widened aperture (`nonuniform_grids.md`, Section 7).

7 Discrete symbols

The discrete symbol of the scheme. The discrete scheme has a discrete symbol in $q \in (-\pi/\Delta\eta, \pi/\Delta\eta]$. For frozen coefficients it is manuscript (18),

$$L_h(q) = \nu - \frac{4}{\tau^2 r^2} \frac{\sigma}{1 - 4\alpha\sigma}, \qquad \sigma = \sin^2 \frac{q\Delta\eta}{2},$$

which replaces $b/\beta^2 - 1 = \nu - q^2/(\beta r)^2$ by a periodic function that agrees with it to $O((q\Delta\eta)^6)$ for $\alpha = 1/12$. The step's symbol is $R(L_h(q))$, and the dispersion error (19) is the distance between this and the exact symbol of the previous section.

Local equivalence. Manuscript (20) says that the full discrete symbol depends on the metric only through the products $h_\xi\Delta\xi$, $h_\eta\Delta\eta$ and β/h_ξ. That is a statement about symbols: in physical variables, the discrete symbol of a distorted cell equals the flat discrete symbol on the effective grid.

Variable coefficients. The weighted second difference with half-node values $r_{j\pm1/2}$ is the discrete analogue of the operator form in (2). Its discrete Weyl symbol has the same structure as (5): no first-order part, and a zeroth-order correction that converges to W as $\Delta\eta \to 0$. This is the discrete form of the remark that W is carried implicitly.

The vertical step rule. The error of L_h is a function of $q\Delta\eta = k\sin\theta \cdot h_\eta s\Delta\eta$, by (6) with $h_\eta \to h_\eta s$. Bounding it at every level requires $\max_\xi h_\eta s\,\Delta\eta \le \Delta z_{\text{phys}}$. This is the interior stretching rule (`nonuniform_grids.md`, Section 6). The manuscript's uniform rule bounds the same quantity by the cruder $h_+ \max r$.

8 The shift map in symbols

Write the shift-map equation as $\psi_{xx} - 2s\psi_{x\zeta} + (1+s^2)\psi_{\zeta\zeta} + k^2\psi = 0$, with $s = T'$. Its symbol in (κ, q), the duals of (x, ζ), is $-\kappa^2 + 2s\kappa q - (1+s^2)q^2 + k^2$, so the outgoing root is

$$\kappa = sq + \sqrt{k^2 - q^2}. \tag{16}$$

- *The obstruction.* The one-way symbol has a part sq that is odd in q. No function of a single symmetric second-order vertical operator without a first-order term has a symbol with an odd part. This is the symbol form of "no function of a single vertical operator is the exact propagator" (manuscript, Section III).

- *The gauge.* The gauge (23) multiplies by $e^{ik_0 s\zeta}$, which translates the dual variable, $q = q' + k_0 s$. Combined with the range phase it gives

  $$\kappa' = sq' + \tfrac{k_0 s^2}{2} + \sqrt{k^2 - (q' + k_0 s)^2} - k_0.$$

  If the square root is truncated at second order, the narrow-angle (NA) symbol, the terms linear in q' and the constants cancel and $\kappa' = -q'^2/(2k_0)$ exactly: the BT transformation is exact for the narrow-angle PE. The higher terms of the root involve $(q' + k_0 s)^4, \ldots$ and do not cancel.

- *The floor.* A wide-angle PE uses the symbol $k_0[f(-q'^2/k_0^2) - 1]$, an even function of q', while the exact κ' is not even beyond second order. The difference is the floor (25). It is a mismatch of symbols, independent of the grid, so no refinement removes it.

- *The orthogonal map.* The exact one-way symbol is $\sqrt{b_0}$, a function of the symbol of the single operator B, so no floor arises.

9 The top boundary

Continuous exterior. Above the transition height, $h_\xi = h_\eta = 1$ and the exterior is invariant in x. In the Fourier variable κ (dual to x), the field there is $e^{i\sqrt{k^2-\kappa^2}\,z}$, so the transparent condition is

$$\partial_z \psi = i\ \mathrm{Op}_x^w\Big(\sqrt{k^2-\kappa^2}\Big)\,\psi.$$

This is a pseudodifferential operator in the evolution variable x, hence a convolution in range. That is why the exact NLBC (5) carries the whole history. Its symbol has branch points at $\kappa = \pm k$, i.e. at vertical wavenumber $k_z = 0$: waves that graze the top.

Discrete exterior. In the scheme, the z-transform variable t plays the role of $e^{i\kappa\Delta x}$, and the scalar exterior ratio $R(\lambda) = u_{N-1}/u_N$ is the discrete symbol of the condition. Here $\lambda = \beta^2(1+\sigma)$ is the spectral variable of the β-free operator A (`nonuniform_grids.md`, Section 8). $R(\lambda)$ has the discrete branch point where the two roots $\mu = e^{\pm i k_z \Delta z}$ meet, again at $k_z = 0$.

Local conditions are rational approximations. A boundary condition that is local in range, such as an absorbing layer or a complex-step layer of M rows, has an exterior ratio that is a rational function of λ with finitely many poles. A rational function cannot approximate a square root uniformly up to its branch point:

- on the band $k_z\Delta z \in [\theta_{\min}, \theta_{\max}]$ the best error decays geometrically in M, with a rate that depends on $\log(\theta_{\max}/\theta_{\min})$ [8, 9];
- as $\theta_{\min} \to 0$ the rate goes to zero.

In the continuous picture the layer's complex thickness Λ gives a reflection of about $e^{-2k_z \operatorname{Im}\Lambda}$, which tends to 1 as $k_z \to 0$ for any finite layer. The complex layer of the code is the optimal-grid construction of [9, 10] applied to the Numerov exterior. The observed rows (8, 12, 16 and 34 rows for $5\cdot10^{-3}$, $2\cdot10^{-4}$, 10^{-5} and 10^{-6} over 0.5–30°) show this geometric rate.

Why the layer serves every class. For per-class β_k, Padé chains and range steps, the exterior equation of every substep is $(A-\lambda)w = 0$ with the same A. A condition expressed as a function of λ, i.e. of the symbol of A alone, is therefore valid for all of them. The convolution NLBC is not: it is a function of t, and the relation between t and λ changes with the class.

10 Summary of orders

term	symbol	order
dispersion relation	$b_0 = k^2 h_\xi^2 - q^2/r^2$	1
first-order (transport) part of B	0 (Liouville gauge)	—
V, W	$(r^{1/2})_{\xi\xi}/r^{1/2}$, $r_\eta^2/(4r^4)$	ε^2
transport term of the one-way equation	$\frac{i}{4} b_\xi / b$	ε (removed by $B^{1/4}$)
commutators, second-order WKB	(9), $i\{b, b_\xi\}$	ε^2
flux pairs at neighbouring columns	$\{b_n, b_{n+1/2}\}$	$\varepsilon\,\Delta\xi$
Bremmer coupling $\mathcal{R}$	$b_\xi/(4b)$	ε
slaved part (local)	$-\rho^2/(2\lambda_0)$, part of λ_2	ε^2
genuine backscatter	$\int \rho\, e^{2i\Phi} d\xi$	ε^∞ $(e^{-2kh\cos\theta\,\delta})$
staircase reflection loss (avoided by flux pairs)	$\rho^2 \Delta\xi^2$ per step	$\varepsilon^2 \Delta\xi$ total
rational step vs exponential	$R(b/\beta^2 - 1) - e^{i\Delta\xi(\sqrt{b}-\beta)}$	set by the optimizer
Numerov symbol	$L_h(q) - (b/\beta^2 - 1)$	$(q\Delta\eta)^6$
shift map, wide angle	odd part of $sq + \sqrt{k^2 - q^2}$	$O(1)$, grid independe
local top condition	rational approximation of $\sqrt{\cdot}$ near $k_z = 0$	geometric in M, not

References

[1] M. Zworski, *Semiclassical Analysis*. Providence, RI: AMS, 2012.

[2] M. Dimassi and J. Sjöstrand, *Spectral Asymptotics in the Semi-Classical Limit*. Cambridge Univ. Press, 1999.

[3] B. Helffer and D. Robert, "Calcul fonctionnel par la transformation de Mellin et opérateurs admissibles," *J. Funct. Anal.*, vol. 53, pp. 246–268, 1983.

[4] L. Fishman and J. J. McCoy, "Derivation and application of extended parabolic wave theories," *J. Math. Phys.*, vol. 25, pp. 285–308, 1984.

[5] M. V. de Hoop, "Generalization of the Bremmer coupling series," *J. Math. Phys.*, vol. 37, pp. 3246–3282, 1996.

[6] H. Bremmer, "The W.K.B. approximation as the first term of a geometric-optical series," *Comm. Pure Appl. Math.*, vol. 4, pp. 105–115, 1951.

[7] F. V. Atkinson, "Wave propagation and the Bremmer series," *J. Math. Anal. Appl.*, vol. 1, pp. 255–276, 1960.

[8] D. J. Newman, "Rational approximation to $|x|$," *Michigan Math. J.*, vol. 11, pp. 11–14, 1964.

[9] D. Ingerman, V. Druskin and L. Knizhnerman, "Optimal finite difference grids and rational approximations of the square root. I. Elliptic problems," *Comm. Pure Appl. Math.*, vol. 53, pp. 1039–1066, 2000.

[10] S. Asvadurov, V. Druskin, M. N. Guddati and L. Knizhnerman, "On optimal finite-difference approximation of PML," *SIAM J. Numer. Anal.*, vol. 41, pp. 287–305, 2003.